Department of Microbiology Quarterly Newsletter

September 2025



Quarterly newsletter highlighting the Department of Microbiology at NYU Langone

Fall Back Into Focus

As summer winds down and the fall semester begins, we're back with our second quarterly newsletter! In this issue, you'll find upcoming seminars and conferences, research highlights, new faces (including some tiny ones), a few smiles for your next incubation break, and even a chance to name our newsletter! If you didn't get to share your good news this round, just click the button and "Share an Update" with your Content Ambassador for inclusion in the next issue!

Here's to a strong start to the year!

~Your Newsletter Editors, Judy & Emily

Share an Update

Section Shortcuts

A Note from the Chair
Vote to Name Our Newsletter!
Upcoming Seminars
Conferences
Department Updates
Spotlight on Success
Publications & Presentations
Personal Milestones
New Faces
Fond Farewells
A Tribute: David Baltimore (1938-2025)
Just For Fun! 😄
Errata
Are We Missing Anyone?

A Note from the Chair



Despite current events that continue to undermine U.S. science, this past year has proven to be one of growth and renewal for our Department of Microbiology.

By recruiting new faculty and fostering a greater sense of community, we have seen measurable improvements in this year's engagement survey. And while there is still work to be done, what matters is that we are moving from being a collection of individual labs to becoming a single cohesive Department; one where seminars, defenses, and shared celebrations are not just events, but expressions of who we are.

History reminds us that science funding rises and falls with world events, often leaving our field vulnerable to volatility and uncertainty. While we cannot control these political storms, we can control the strength of our community and the pride we take in our work. That is how we turn uncertainty into momentum, and how we ensure that our department continues to thrive, regardless of what the world throws our way.

I look forward to sustaining our momentum in the year ahead and to the shared rewards of what we can create together.

Vote to Name Our Newsletter!

Thanks to everyone who voted last quarter to narrow down our list for a new newsletter name. We have a three-way tie!

Click the button to VOTE for the one you like!

VOTE
to break the tie!

DEADLINE: September 30th

Upcoming Seminars

INTERNAL SEMINARS

Annual Department of Microbiology Address
Benjamin tenOever, PhD

September 12th

Departmental Address:
11:30 AM - 12:30 PM
Science Building
Room 103

Reception:
4:00 PM - 6:00 PM
ACLS West 5th Floor



RECURRING MEETINGS

Departmental Seminar Series

NEW TIME!

Fridays at 11:30 AM
Science Building
Room 103

Departmental Seminar Series Calendar

Virology Group Meeting

10:30 AM - 11:30 AM
Every other Tuesday
ACLS West 5th Floor
Conference Room 508

Virology Consortium Meeting Schedule

UPCOMING THESIS DEFENSES

Nora Herzog

September 17th
3:00 PM
Alumni Hall A

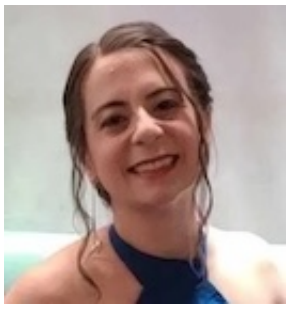
Zoom link:
https://nyulangone.zoom.us/j/9877039602
pwd=0cZl6b9dfPq4wbReGOjbbjJPnZb2n



Rebecca Plessel

October 8th
1:00 PM
Smilow

Zoom link: TBA



Rebecca Plessel, a graduate student in the Mohr/Wilson group, will be defending her thesis in October. Title and details to come.

OTHER SEMINARS OF INTEREST

The Vaccine Dinner Club

Online (link provided with registration confirmation)

Agenda: (times in EST)
6:30 PM – Meeting begins
8:00 PM – Q & A

Join VDC

What began as a small monthly gathering over dinner and science has grown into one of the largest communities of vaccinophiles in the world. Today, the Vaccine Dinner Club (VDC) connects over 5,000 members worldwide—from students and researchers to public health leaders and industry partners—dedicated to advancing vaccine science and building a supportive scientific community. Each month, invited luminaries in the field share their work as featured speakers.

Membership is free and meetings are open to all, with live access via Zoom and recordings available for registered participants.

Past speakers include: Michael Osterholm, Walter Orenstein, Barney Graham, Florian Krammer, Peter Hotez, Paul Offit, William Foege, Rafi Ahmed, and so many more.

In honor of Dr. Edward Jenner, "Uno," an American domestic cow who collaborates in vaccine research with VDC member Amelia R. Woolums, DVM, PhD, Department of Large Animal Medicine at the University of Georgia, is the official mascot of the VDC. The term vaccination comes from the Latin *vacca* (for cow) and was invented by Dr. Jenner after deriving the first immunization for smallpox from cowpox matter he got from the hand of milkmaid Sarah Nelmes.



Conferences

IMARI 2026

January 28 - 30, 2026: Las Vegas, NV
The American Society for Microbiology and the Infectious Diseases Society of America are excited to announce the launch of the Interdisciplinary Meeting on Antimicrobial Resistance and Innovation (IMARI). This inaugural conference will serve as a one-of-a-kind forum for collaboration and exploration around the latest advances in antimicrobial drug discovery and development.

Immunology 2026

April 15 - 19, 2026: Boston, MA
109th annual meeting of the American Association of Immunologists (AAI) held at the Thomas M. Menino Convention & Exhibition Center (MCEC) in Boston, MA. Abstract submission opens from Sept 3 - Oct 15.

ASM Microbe 2026

June 4 - 8, 2025: Washington, DC
American Society for Microbiology annual meeting with a new format that brings together the full breadth of microbiology through 3 distinct, yet interconnected meetings, all under one roof:

- ASM Health
- ASM Applied & Environmental Microbiology
- ASM Mechanism Discovery

Microbes in Microbiomes 2026

February 18 - 20, 2026: Miami, FL
The inaugural Microbes in Microbiomes conference will showcase advances in microbiome science, from structure to function, with a focus on the individual microbes driving key traits. Topics span human, animal, plant, and environmental health, with an emphasis on species-level insights and their interactions within complex communities.

ASGCT 2026

May 11 - 15, 2026: Boston, MA
29th annual meeting of the American Society of Gene and Cell Therapy (ASGCT), held at the Boston Convention and Exhibition Center in Boston, MA.

Submission details to be announced.

GRC 2026: Microbial Toxins and Pathogenicity

July 12 - 17, 2026: Waterville Valley, NH
The 2026 Gordon Research Conference (GRC) on Microbial Toxins and Pathogenicity and its companion Gordon Research Seminar (GRS) will feature research on the fundamental mechanisms of host-microbe/toxin interactions and bacterial pathogenesis.

Invited Keynote Speaker: Heran Darwin

FOCIS 2026

June 9 - 12, 2026: San Francisco, CA
26th Annual Meeting of the Federation of Clinical Immunology Societies, held at the San Francisco Marriott Marquis in San Francisco, CA.

ASV 2026

July 27 - 30, 2026: Minneapolis, MN
45th annual meeting of the American Society for Virology, hosted by the University of Minnesota.

Abstract submission opens December 1st.

IUMS 2026

November 4 - 6, 2026: Lisbon, Portugal
The International Union of Microbiological Societies (IUMS) connects global microbiological societies to promote sustainable solutions for infectious agent control while preserving microbial diversity and planetary health. The IUMS Biennial Congress unites all the main three disciplines of microbiology—Bacteriology and Applied Microbiology, Virology, and Mycology and Eukaryotic Microbiology—in one place.

Abstract submission deadline April 29th.

ASTMH 2026

November 9 - 13, 2026: Toronto, Canada
The American Society of Tropical Medicine and Hygiene (ASTMH) annual meeting will be held at the Metro Toronto Convention Centre in Toronto, Canada. There are options available to livestream certain sessions for anyone interested who cannot attend in person.

Department Updates

New Initiatives

Collaboration Watch

Heran Darwin shares that her lab is initiating a collaboration with Da-Neng Wang (Biochem/Cell Bio) and Alejandro Pironti—stay tuned for new insights and discoveries emerging from this partnership!

Departmental Wiki

We are working towards developing a Departmental Wiki page in the months ahead. Stay tuned.

Equipment and Facilities

The tenOever lab has a new home!

The tenOever Lab has relocated to the 4th floor as of late August. Be sure to visit them in their new space.

New equipment on the horizon?

Not final yet, but the Department may be receiving some new equipment pending OSR approval when the Department moves into its designated space on floors 3-5 in ACLS-W. 🤞🏻

Spotlight on Success

PI News

Heran Darwin, PhD





Congratulations to Heran Darwin on her induction into the National Academy of Sciences (NAS) this past April! As part of the ceremony, she had the honor of signing the same membership book once signed by Thomas Edison and other scientific luminaries—joining a tradition that celebrates a legacy of groundbreaking discovery.

Dr. Darwin, whose career has focused on how *M. tuberculosis* shapes the immune system of its host, was elected to the NAS this year as one of just 120 new members (including 24 international), and is among only 23 NYU faculty ever honored.

Video: NAS 162nd Annual Meeting—Research Briefings: K. Heran Darwin

Photo credit: National Academy of Sciences

Trainee News

Alberto Canfrán-Duque, PhD



American Heart Association.

Career Development Award

Alberto Canfrán-Duque, a Research Scientist in the Schneider lab, has been awarded the prestigious Career Development Award from the American Heart Association. This grant supports highly promising health care and academic professionals in the early stages of their careers. It encourages the pursuit of innovative research questions and pilot studies that generate preliminary data and provide essential training for long-term success as a research scientist.

The funded project, “Mechanisms of mRNA Translation Reprogramming in Macrophage Plasticity and Atherosclerosis Progression,” will explore how changes in the mRNA translation machinery drive macrophage diversity and the development of atherosclerosis. By integrating advanced translational profiling techniques, this work aims to uncover novel molecular mechanisms and potential therapeutic targets for cardiovascular disease.

Sheya Xiao Ma, PhD



Congratulations to Sheya Ma, a graduate student in the Weiser lab, who successfully defended her thesis on June 26th!

Thesis title: "Inhibitory effect of capsule on natural transformation of Streptococcus pneumoniae"

Sydney Pimentel



Sydney Pimentel, an MD/PhD student in the tenOever lab, successfully passed her qualifying exam in July!

Margaux Sica, PhD

Adam Silhavy



Congratulations to Margaux Sica, a graduate student in the Rodriguez lab, who successfully defended her thesis on July 17th!

Thesis title: "The Broadly Reactive Antibody Response to Malaria."



Adam Silhavy, a graduate student in the Mohr/Wilson group, successfully passed his qualifying exam in May!

Alumni News

Gavyn Bee, PhD



Gavyn Bee, a former graduate student in the Weiser lab, was awarded the 2025 Helen Hay Whitney Postdoctoral Fellowship at the University of Pennsylvania.

Samuel Becker, PhD



Sam Becker, a former graduate student in the H Darwin lab, accepted an Assistant Professor position at Stony Brook University, and is starting his lab there this fall.

Shoshana Kahne, PhD



Shoshana Kahne, a former graduate student in the H Darwin lab, received a Jane Coffin Childs fellowship at UW Seattle, working with Alex Meeske.

Ifrah Shahi, PhD



Ifrah Shahi, a former graduate student and postdoc in the Ratner lab, has accepted a tenure track Assistant Professor position at Bates College. She will open her lab and begin teaching there this fall!

Thank you to all our Postdocs!

Since 2009, the National Postdoctoral Association (NPA) has sponsored National Postdoc Appreciation Week to recognize the significant contributions that postdoctoral scholars make to U.S. research and discovery. Thank you, Microbiology postdocs, for everything you do to make our department thrive!



Publications & Presentations

H. Darwin Lab

Methylglyoxal is an antibacterial effector produced by macrophages during infection
Anaya-Sanchez, et al. **Cell Host & Microbe.**

Garabedian & Dittmann Labs

In-silico docking platform with serine protease inhibitor (SERPIN) structures identifies host cysteine protease targets with significance for SARS-CoV-2.
Rodriguez Galvan, et al. **Preprint in press with EMBO.**

Khanna Lab

Vagal TRPV1+ sensory neurons protect against influenza virus infection by regulating lung myeloid cell dynamics.
Almanzar, et al. **Science Immunology.**
*also featuring tenOever lab alum: Daisy Hoagland

Macrophages: sentinels, warriors, and

Ortigoza Lab

Stacey Bartlett gave a talk and other members of the lab attended and presented at the American Society for Virology (ASV) meeting (July 14-17, 2025 in Montréal, Canada).



Stapleford Lab

Nicole Rondeau presented a poster on her work with La Crosse virus at the Negative-strand RNA Virus meeting (June 22-27, 2025 in Montpellier, France) and won the prize for her section!

healers. (Review)
Bernier, Bartnicki, and Khanna. **Human Molecular Genetics.**

Nerve-associated macrophages control adipose homeostasis across lifespan and restrain age-related inflammation.
Gonzalez-Hurtado et al. **Nature Aging.**

Mohr/Wilson Group

De novo assembly of RNA m6A modification factors into viral genome-associated nuclear bodies drives HCMV RNA accumulation.
Grande, et al. **Cell Rep.**

Denise Ohnezeit presented her work at the 49th Annual International Herpesvirus Workshop (July 26-30, 2005 in Berlin, Germany)

tenOever Lab

Leon Hsieh, **Morgane Baldaccini**, and **Ryla Cantergiani** gave talks and **Sydney Pimentel** gave a flash talk and presented a poster at the American Society for Virology (ASV) meeting (July 14-17, 2025 in Montréal, Canada).

Weiser Lab

Inhibitory effect of capsule on natural transformation of *Streptococcus pneumoniae*.
Ma, et al. **mBio.**

Personal Milestones



Congratulations to **Lucia Carrau** (tenOever lab), who gave birth to Facundo Ignacio Rivera Carrau on June 8th!

Lucia says, "Everything is going great! Baby Facundo is a sweetheart and I'm enjoying this new life a lot!"



Ryla Cantergianni (tenOever lab) got married on April 25th at Sure Thing Chapel in Las Vegas, NV. She reports, "It was a nice, intimate ceremony that was just with our immediate



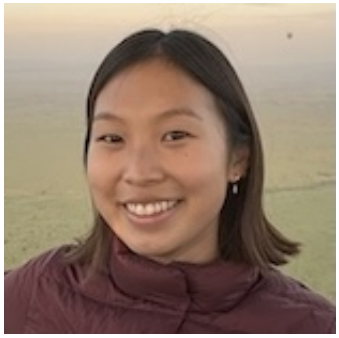
Congratulations to **Phuong Tran** (H. Darwin lab) on the birth of Oliver Mahling on June 25th!

family. Our theme was “Lucky in Love” because it was appropriately corny. The entire ceremony was 9 minutes long (my father-in-law timed it as a joke because I promised it would be under 15min)."

Congratulations and much happiness to Ryla and Carli! 💖

New Faces

Phoebe Chen



Phoebe Chen, a graduate from the University of California Los Angeles, is a new graduate student in the Mohr/Wilson group.

Molly Durawa



Molly Durawa, a computational biology and history graduate from Barnard College and Columbia University, has just started as a Research Technician in the Stapleford lab.

Ryan Gado



Ryan Gado, a second year graduate student,

Abigail Korenek



Abigail Korenek is a new MD/PhD student

who started his rotation in the H. Darwin lab in April, has officially joined the lab!

Heemaja Mewada



Heemaja Mewada joined the Khanna lab as a graduate student. She earned her BS in Biochemistry from UNC Chapel Hill and worked as a research tech in the lab of Laurie Glimcher at Dana Farber.

Nikunjkumar Vagadiya



Nikunjkumar Vagadiya is graduate student in Chemistry at the NYU Washington Square Campus in Kent Kirshenbaum's lab, and is collaborating with the Garabedian lab on developing and characterizing AR-targeting peptoid.

in the tenOever lab.

Leigh Miller, PhD, MSPH



Leigh Miller is a new postdoctoral fellow in the Khanna lab. She holds a Master's of Science in Public Health from Tulane University, and recently earned her PhD from the University of Pittsburgh working on host immune responses to influenza virus and MRSA in the lung.

Eliza Zaroff



Eliza Zaroff, a graduate of Georgetown University, will start in September as a new Research Associate in the A. Darwin lab.

Fond Farewells

Sedem Apedo

Shirley Braspenning, PhD

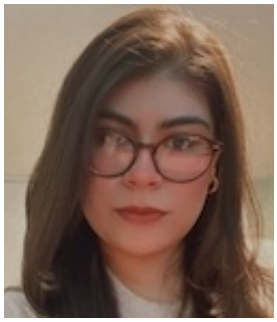


Sedem Apedo, an NYU undergraduate in the tenOever lab, graduated with honors this past Spring, and has left the lab. We wish her the best in her future endeavors!



Shirley Braspenning, a postdoctoral fellow in the Mohr/Wilson group, departed NYU to begin a new postdoc in Gerald McInerney's lab at the Karolinska Institutet.

Lizeth Chicas



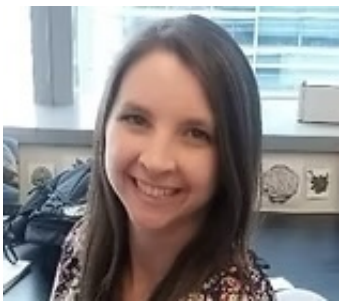
Lizeth Chicas, lab Technician and former master's student in the Rodriguez lab, left the lab this summer and now holds an adjunct faculty position at Seton Hall University.

Hannes Eichner, PhD, MSc



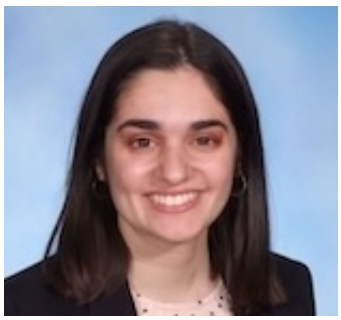
Hannes Eichner, a postdoctoral fellow in the Weiser lab, left at the end of May.

Kristen Lokken-Toyli, PhD



Kristen Lokken-Toyli, Senior Research Scientist in the Weiser lab, left at beginning of May to become a Research Associate in the Knoop lab at the Mayo Clinic in Rochester, MN.

Francesca Serraino



Francesca Serraino, a Research Technician in the Garabedian lab, left to pursue a Master's degree in biotechnology at the University of Pennsylvania.

Julia Sproch, PhD



Julia Sproch (Ratner lab) defended her thesis "Elucidating the Mechanism of Host Cell Lysis by *S. aureus* Leukocidin HlgAB" in early June. She is still living in NYC where she is now working as a Medical Writer at Real Chemistry.

A Tribute: David Baltimore (1938-2025)

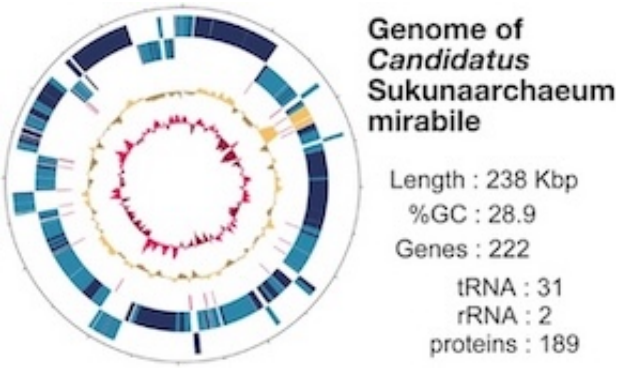


David Baltimore, PhD, Nobel Prize–winning molecular biologist, passed away on September 6th at his home in Woods Hole, MA. Best known for his discovery of reverse transcriptase—a breakthrough that reshaped our understanding of how genetic information flows and led to critical insights about HIV—he made transformational contributions to virology, molecular biology, and immunology. He was awarded the Nobel Prize for this discovery at just 37, and he went on to lead institutions such as MIT, Rockefeller University, and Caltech. He was an early advocate for investments into HIV/AIDS research, tirelessly urging the scientific community to confront attention on the newly emerged virus while guiding and mentoring generations of scientists. His brilliance, advocacy, and mentorship will leave an enduring legacy.

Read the full article in the NY Times.

Just for Fun! 😄

Microbiology in the News



Microbe with bizarrely tiny genome may be evolving into a virus

Scientists have identified a strange new microbe that blurs the line between viruses and cellular life. Provisionally named after a small-statured deity in Japanese mythology, Sukunaarchaeum was found by chance by researchers at the University of Tsukuba while trying to sequence the DNA of the dinoflagellate *Citharistes regius.* The tiny 238,000 bp circular DNA genome of Sukunaarchaeum encodes just 189 protein-coding genes, almost exclusively dedicated to DNA replication, transcription, and translation. With virtually no metabolic pathways of its own, it depends almost completely on its host for survival, much like a virus. However, unlike viruses, it can replicate its own genetic material. Researchers believe Sukunaarchaeum may represent a rare evolutionary intermediate, offering insight into how viruses might have first evolved.

The preprint:
A cellular entity retaining only its replicative core: Hidden archaeal lineage with an ultra-reduced genome.
Harada et al. bioRxiv (preprint)

2025 Dance Your PhD Winners



University of Helsinki food scientist, Sulo Roukka, won Science's "Dance Your Ph.D." contest with a performance illustrating how food compounds like capsaicin and menthol trigger different sensations, a phenomenon known as chemesthesis. The project, which won in the Overall and Chemistry categories, highlights how sensory effects shape the experiences and memories of food, with potential applications for developing plant-based products and nonalcoholic drinks.

Watch all the winning videos here.

Fall-themed Food Ideas



Jack-o'-lantern stuffed peppers



Apple-Pumpkin Upside-Down Cake



Candy Corn Cocktail



2025 Ig Nobel Prizes

Thursday, September 18th
6:00pm

Boston University (George Sherman Union ballroom)

Available by webcast.

What is Improbable Research?
Real research, about anything and everything, from everywhere that makes people LAUGH, then THINK.

This year, TEN new Ig Nobel Prize winners will be introduced. The prizes will by handed to them by a gaggle of bemused Nobel laureates. An audience will shower them with paper airplanes.

A sample of winning research from last year:
BIOLOGY PRIZE for exploding a paper bag next to a cat that's standing on the back of a cow, to explore how and when cows spew their milk.

CHEMISTRY PRIZE for using chromatography to separate drunk and sober worms.

MEDICINE PRIZE for demonstrating that fake medicine that causes painful side-effects can be more effective than fake medicine that does not cause painful side-effects.

Share Your Favorite Local Coffee Shop

☕✨ Love your local coffee spot? Whether it has the perfect latte or the coziest corner to get work done, share your caffeine secrets with your microbiology colleagues! We'll feature everyone's favorites in the next newsletter so we can all discover the best places to sip, study, and recharge!

Submit Your Favorite
Local Coffee Shop

The Petri Dish—where laughs are cultured



Hospitals across the US have reported cases of the dreaded "peek-a-boo virus." 👀

Patients are reportedly transferred to the ICU.

Errata

This section highlights anything we missed in the last newsletter. If you spot an error, just let us know and we'll make sure it's corrected in the next issue.

Congratulations again to Stephen Yeung and Payal Damani-Yokota, co-first authors of this newly published paper! We missed the full citation in our last issue, but now that it's out, here's the link.

- **Nerve and Airway Associated Interstitial Macrophages Mitigate SARS CoV 2 Pathogenesis via Type I Interferon Signaling.** (Yeung, Damani-Yokota, et al. **Immunity**)

Are We Missing Anyone?

If you know anyone in the Micro Department who isn't receiving these emails, please forward this email to them and ask them to click the subscribe button below so we can add them.

Subscribe

NYU Langone Health

Copyright © {FormatDate(Today;"yyyy")} NYU Langone Health. All rights reserved.

Our mailing address is NYU Langone Health • 550 First Avenue • New York, NY 10016